



Progressive Education Society's
Modern College of Engineering, Pune-05.

(An Autonomous Institute Affiliated to Savitribai Phule Pune

University) Department of T.Y. B. Tech [AIDS]

Assignment No: 02

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Section: A
Roll Number: 36058
Subject : Computer Vision

Title / Problem Statement:

Show translation, rotation, and scaling of an image using mathematical foundations such as matrices and coordinate transformation principles in Geometric Transformations.

Program Input :

```
import cv2  
  
import numpy as np  
  
import matplotlib.pyplot as plt  
  
# Load the image
```

```
image = cv2.imread('BMW.jpeg')  
  
# Convert BGR to RGB for Matplotlib  
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)  
  
  
# Get image dimensions  
rows, cols = image_rgb.shape[:2]  
  
  
# 1. Translation  
  
# Shift 100 pixels right and 50 pixels down  
tx, ty = 100, 50  
translation_matrix = np.float32([[1, 0, tx], [0, 1, ty]])  
translated_img = cv2.warpAffine(image_rgb, translation_matrix, (cols, rows))  
  
  
# 2. Rotation  
  
# Rotate by 45 degrees around the center  
center = (cols / 2, rows / 2)  
angle = 45  
  
# 1.0 is the scale factor (keeping it original size during rotation)  
rotation_matrix = cv2.getRotationMatrix2D(center, angle, 1.0)  
rotated_img = cv2.warpAffine(image_rgb, rotation_matrix, (cols, rows))  
  
  
# 3. Scaling  
  
# Scale down by 0.5x in both directions  
sx, sy = 0.5, 0.5
```

```
scaled_img = cv2.resize(image_rgb, None, fx=sx, fy=sy,  
interpolation=cv2.INTER_LINEAR)
```

```
# Plotting the results
```

```
plt.figure(figsize=(12, 8))
```

```
plt.subplot(2, 2, 1)
```

```
plt.imshow(image_rgb)
```

```
plt.title('Original Image')
```

```
plt.axis('off')
```

```
plt.subplot(2, 2, 2)
```

```
plt.imshow(translated_img)
```

```
plt.title('Translated Image (tx=100, ty=50)')
```

```
plt.axis('off')
```

```
plt.subplot(2, 2, 3)
```

```
plt.imshow(rotated_img)
```

```
plt.title('Rotated Image (45 degrees)')
```

```
plt.axis('off')
```

```
plt.subplot(2, 2, 4)
```

```
plt.imshow(scaled_img)
```

```
plt.title('Scaled Image (0.5x)')
```

```
plt.axis('off')
```

```
plt.tight_layout()
```

```
plt.show()
```

Program Output :

Original Image



Translated Image (tx=100, ty=50)



Rotated Image (45 degrees)



Scaled Image (0.5x)

